

WORKSHEET - NAPLAN Year 9

Number and Algebra Ratios, Scale and Proportion

Teacher: Alex Latham

Exam Equivalent Time: 11.25 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Questions

1. Number and Algebra, NAP-02654

On a country property, 1 acre of land is recommended for every 4 sheep.

How many acres of land would be needed for 16 sheep?

- 4** **6** **16** **64**
- ☐ ☐ ☐ ☐

2. Number and Algebra, NAP-61529

Shelly and Carly collect dolls.

The ratio of the number of dolls Shelly owns compared to Carly is 3:2.

Shelley owns 12 dolls.

How many dolls does Carly own?

- 2** **6** **8** **18**
- ☐ ☐ ☐ ☐

3. Number and Algebra, NAP-79138

In a hospital, there are 40 nurses and 600 patients.

What is the ratio of nurses to patients?

- 2:3** **1:15** **30:2** **1:16**
- ☐ ☐ ☐ ☐

4. Number and Algebra, NAP-02685

A christmas pudding recipe uses 3 cups of sugar for every 4 cups of sultanas.

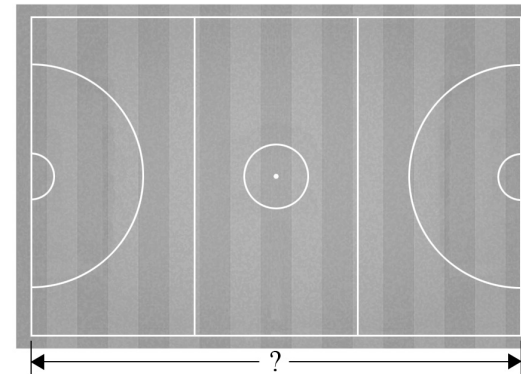
Select the correct combination of sugar and sultanas for this recipe.

- ☐ $\frac{1}{3}$ cup of sugar, $\frac{1}{4}$ cup of sultanas
- ☐ 1 cup of sugar, 1.5 cups of sultanas
- ☐ 1.5 cups of sugar, 2 cups of sultanas
- ☐ 4 cups of sugar, 5 cups of sultanas

5. Number and Algebra, NAP-85080

Libby plays on a hockey field that is 120 metres long.

She makes a scale diagram of the field using a ratio of 1 : 400.



How long should Libby make the scale diagram of the field?

- 3 cm** **30 cm** **90 cm** **300 cm**
- ☐ ☐ ☐ ☐

6. Number and Algebra, NAP-79194

There are 48 people in a meeting room.

The ratio of females to males in the room is 13 to 11.

How many females need to leave the room so that the ratio of females to males in the room is 1 to 1?

7. Number and Algebra, NAP-32158

A school has 360 students.

The ratio of boys to girls is 4 to 6.

How many boys attend the school?

90 **144** **216** **240**

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8. Number and Algebra, NAP-79201

In a class of 28 students, the ratio of boys to girls is 3 : 4.

Two boys and one girl leave the class. No student joins the class.

What is the new ratio of boys to girls?

9. Number and Algebra, NAP-14576

Michelle makes 7.0 litres of a healthy shake to sell at her cafe.

The table below shows the mixture of ingredients that go into the shake.

milk	banana	avocado	water
4 parts	3 parts	5 parts	2 parts

How many litres of milk does Michelle need?

litres